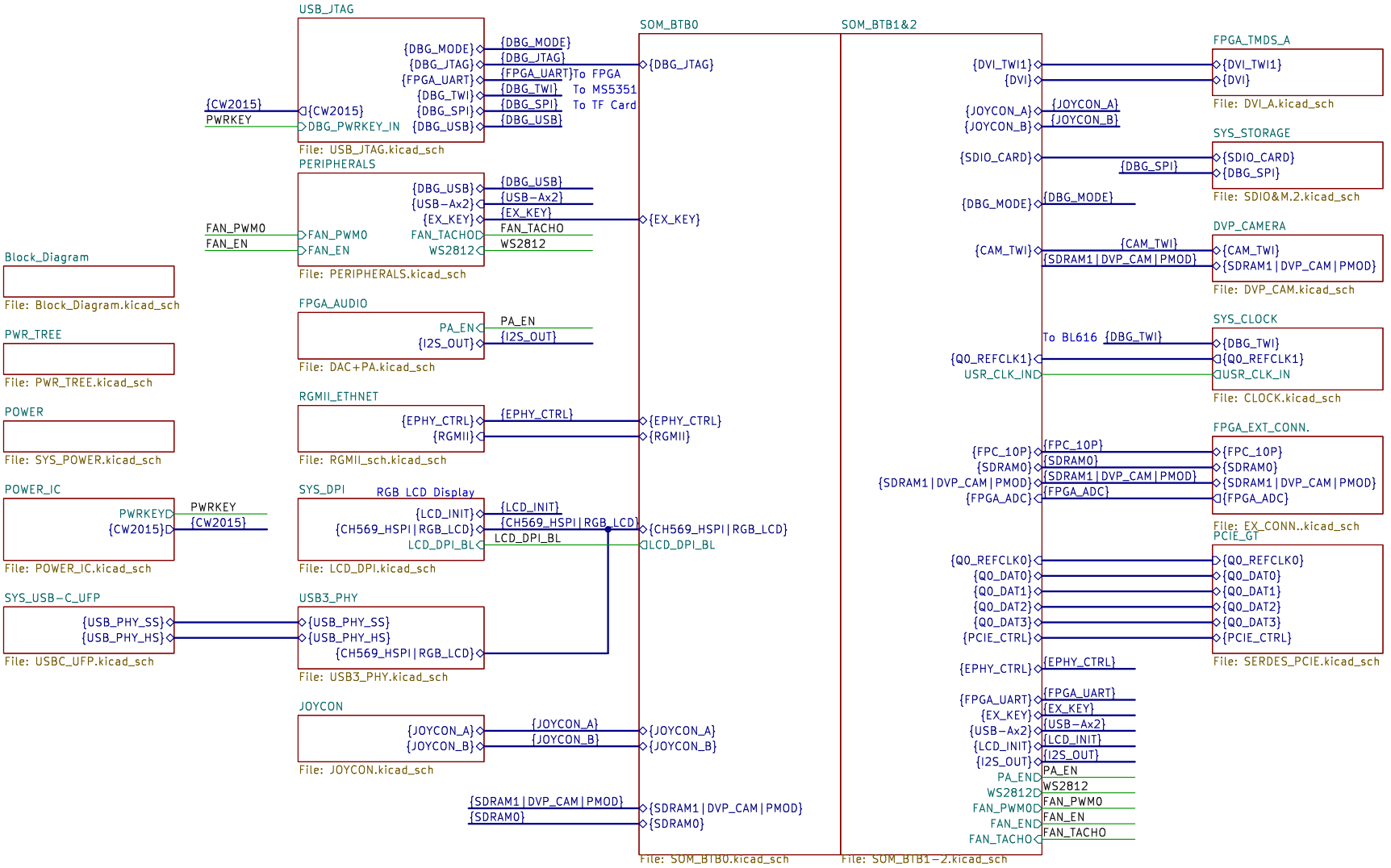
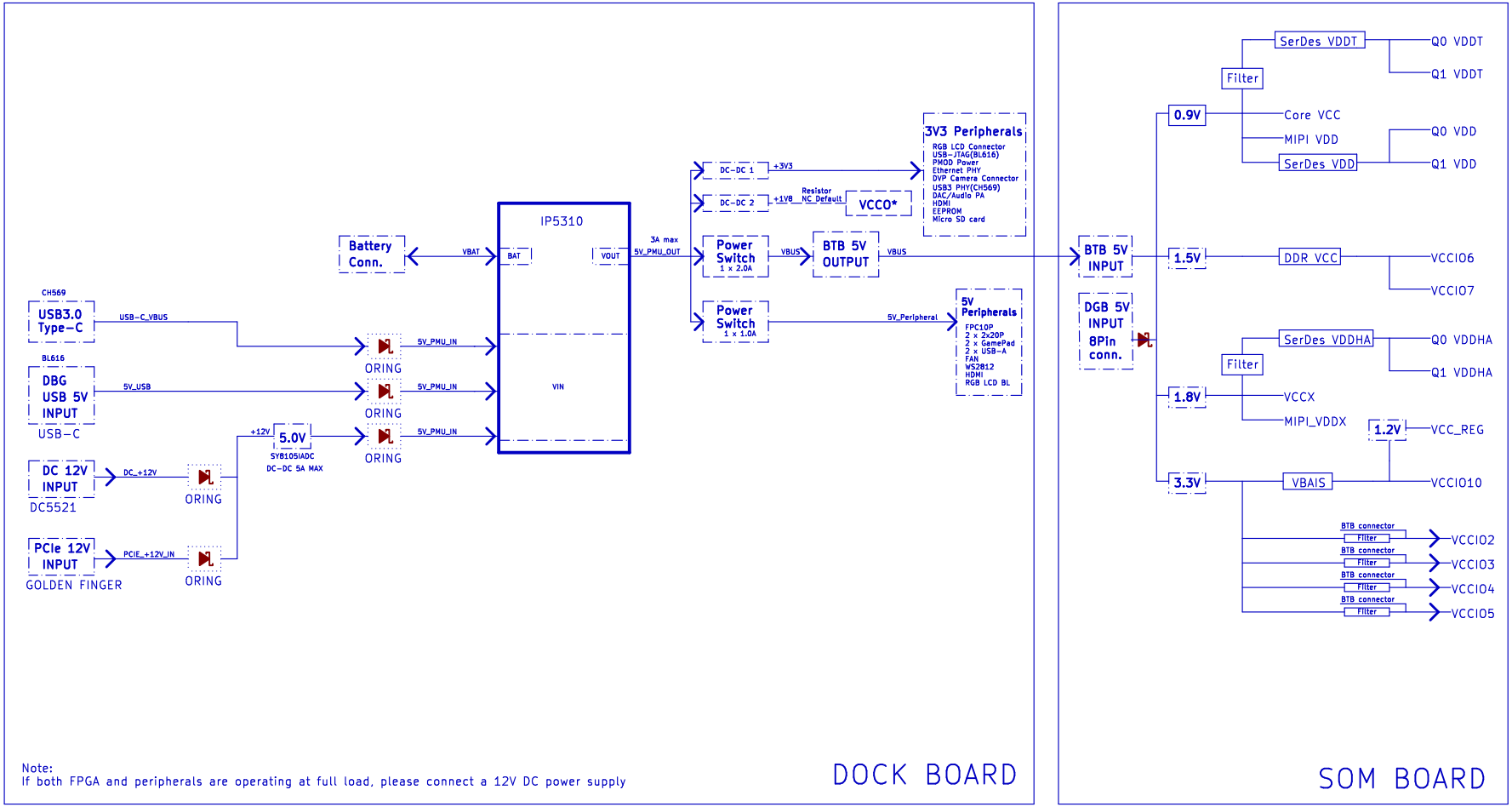


SiPEED

Sheet: /
File: Tang_Mega_60K_Dock_31004.kicad_sch

Title: Tang_Mega_60K_Dock(NEO)

Size: A4	Date: 2025-03-31	**Rev: 1.4**
KiCad E.D.A. 8.0.6		Id: 1/21

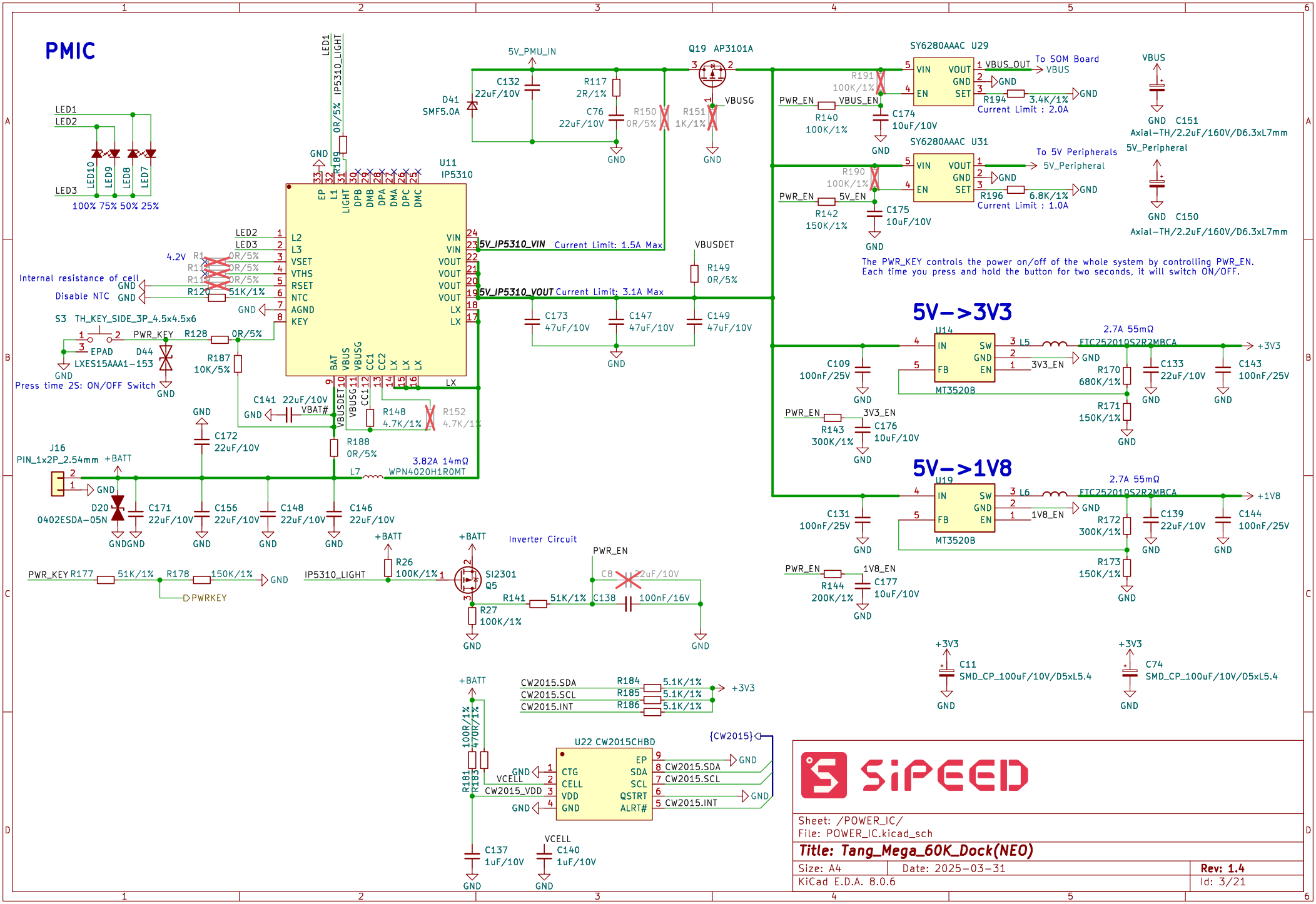


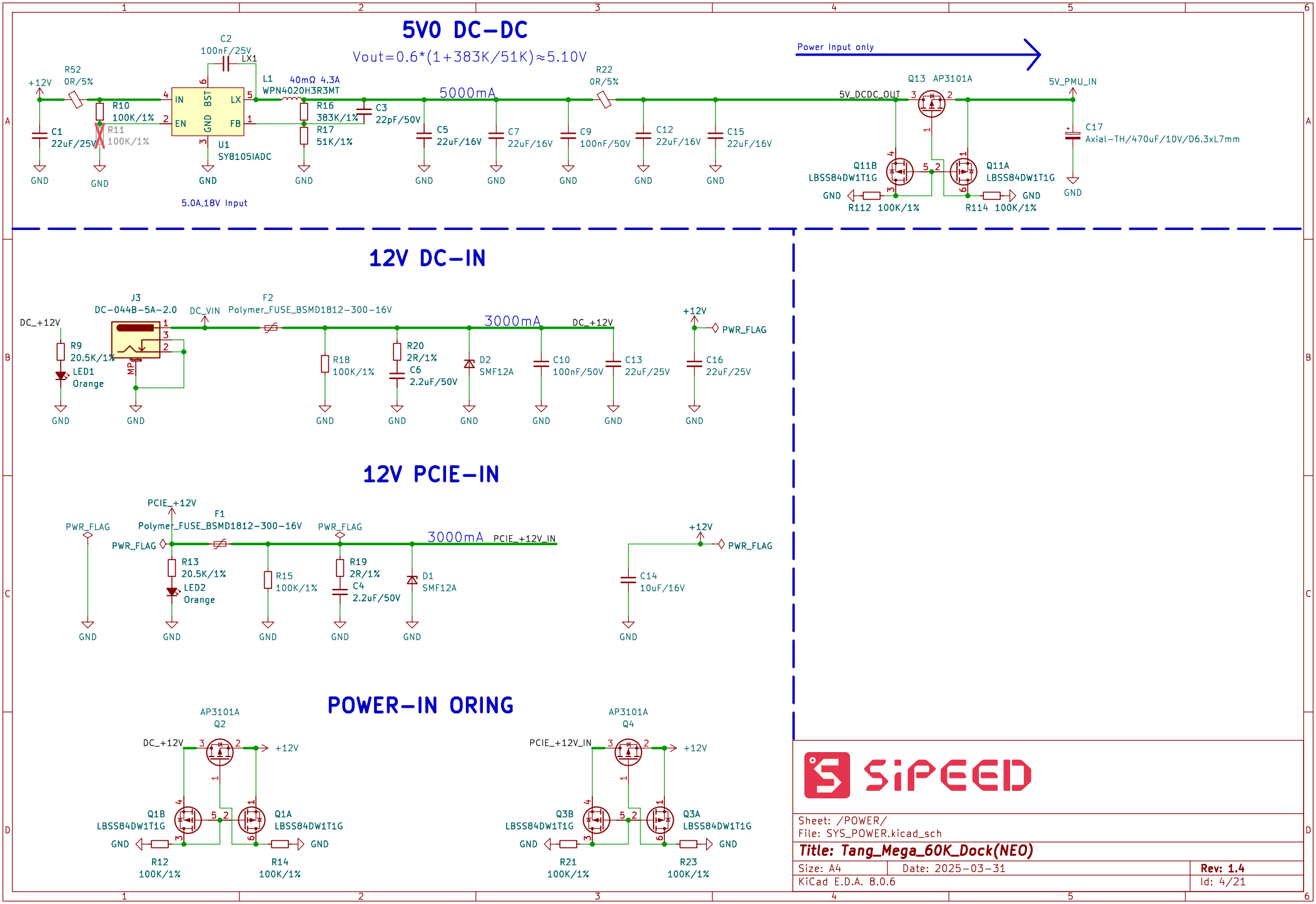
SiPEED

Sheet: /PWR_TREE/
File: PWR_TREE.kicad_sch

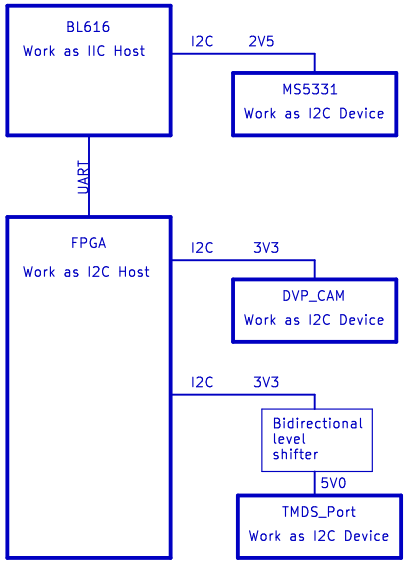
Title: Tang_Mega_60K_Dock(NEO)

Size: A4	Date: 2025-03-31	**Rev: 1.4**
KiCad E.D.A. 8.0.6		Id: 2/21

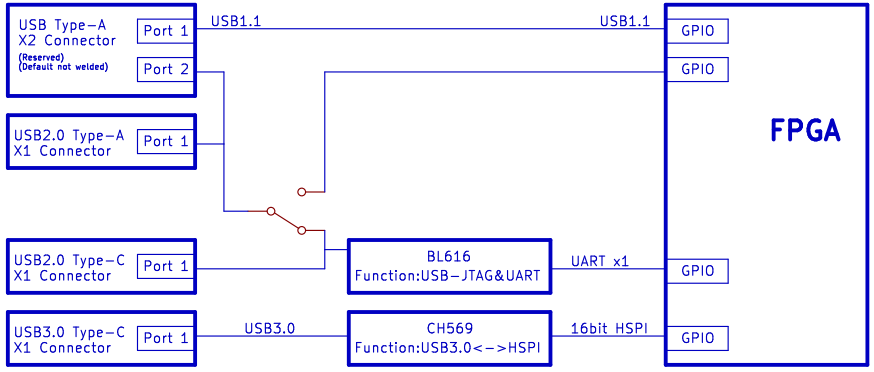

PMIC
LED1
LED2
LED3
LED10
LED9
LED8
LED7
100% 75% 50% 25%
U11
IP5310
5V_PMU_IN
Q19 AP3101A
D41
SMF5.0A
C132
22uF/10V
R117
2R/1%
C76
22uF/10V
R150
0R/5%
R151
1K/1%
VBUSG
SY6280AAAC U29
SY6280AAAC U31
R191
100K/1%
R140
100K/1%
C174
10uF/10V
R194
3.4K/1%
Current Limit : 2.0A
To SOM Board
VBUS
R190
100K/1%
R142
150K/1%
C175
10uF/10V
R196
6.8K/1%
Current Limit : 1.0A
To 5V Peripherals
5V_Peripheral
C151
Axial-TH/2.2uF/160V/D6.3xL7mm
C150
Axial-TH/2.2uF/160V/D6.3xL7mm
5V_IP5310_VIN Current Limit: 1.5A Max
5V_IP5310_VOUT Current Limit: 3.1A Max
VBUSDET
R149
0R/5%
C173
47uF/10V
C147
47uF/10V
C149
47uF/10V
The PWR_KEY controls the power on/off of the whole system by controlling PWR_EN.
Each time you press and hold the button for two seconds, it will switch ON/OFF.
5V->3V3
U14
MT3520B
2.7A 55mΩ
FTC252010S2R2MBCA
C109
100nF/25V
R170
680K/1%
R171
150K/1%
C133
22uF/10V
C143
100nF/25V
R143
300K/1%
C176
10uF/10V
5V->1V8
U19
MT3520B
C131
100nF/25V
R172
300K/1%
R173
150K/1%
C139
22uF/10V
C144
100nF/25V
R144
200K/1%
C177
10uF/10V
C11
SMD_CP_100uF/10V/D5xL5.4
C74
SMD_CP_100uF/10V/D5xL5.4
S3 TH_KEY_SIDE_3P_4.5x4.5x6
D44
LXES15AAA1-153
Press time 2S: ON/OFF Switch
R128
0R/5%
R187
10K/5%
C141
22uF/10V
R148
4.7K/1%
R152
4.7K/1%
R188
0R/5%
L7
WPN4020H1R0MT
3.82A 14mΩ
J16
PIN_1x2P_2.54mm
+BATT
D20
0402ESDA-05N
C171
22uF/10V
C156
22uF/10V
C148
22uF/10V
C146
22uF/10V
C172
22uF/10V
Internal resistance of cell
Disable NTC
R120
51K/1%
4.2V
R189
0R/5%
LED1
IP5310_LIGHT
PWR_KEY
R177
51K/1%
R178
150K/1%
PWRKEY
IP5310_LIGHT
R26
100K/1%
Q5
SI2301
Inverter Circuit
R141
51K/1%
C138
100nF/16V
C8
R27
100K/1%
CW2015.SDA
CW2015.SCL
CW2015.INT
R184
R185
R186
5.1K/1%
U22 CW2015CHBD
{CW2015}
R181
100R/1%
R183
47OR/1%
VCELL
CW2015_VDD
C137
1uF/10V
C140
1uF/10V
SiPEED
Sheet: /POWER_IC/
File: POWER_IC.kicad_sch
Title: Tang_Mega_60K_Dock(NEO)
Size: A4
Date: 2025-03-31
Rev: 1.4
KiCad E.D.A. 8.0.6
Id: 3/21


5V0 DC-DC
Vout=0.6*(1+383K/51K)≈5.10V
Power Input only
+12V
R52
0R/5%
C2
100nF/25V
LX1
L1
40mΩ 4.3A
WPN4020H3R3MT
R10
100K/1%
R11
100K/1%
C1
22uF/25V
IN
BST
LX
EN
GND
FB
U1
SY8105IADC
5.0A,18V Input
R16
383K/1%
R17
51K/1%
C3
22pF/50V
5000mA
C5
22uF/16V
C7
22uF/16V
C9
100nF/50V
C12
22uF/16V
C15
22uF/16V
R22
0R/5%
5V_DCDC_OUT
Q13 AP3101A
5V_PMU_IN
C17
Axial-TH/470uF/10V/D6.3xL7mm
Q11B
LBSS84DW1T1G
Q11A
LBSS84DW1T1G
R112 100K/1%
R114 100K/1%
GND
12V DC-IN
J3
DC-044B-5A-2.0
DC_+12V
DC_VIN
F2
Polymer_FUSE_BSMD1812-300-16V
3000mA
R9
20.5K/1%
LED1
Orange
MP
R18
100K/1%
R20
2R/1%
C6
2.2uF/50V
D2
SMF12A
C10
100nF/50V
C13
22uF/25V
C16
22uF/25V
PWR_FLAG
12V PCIE-IN
PCIE_+12V
F1
Polymer_FUSE_BSMD1812-300-16V
PCIE_+12V_IN
R13
20.5K/1%
LED2
Orange
R15
100K/1%
R19
2R/1%
C4
2.2uF/50V
D1
SMF12A
C14
10uF/16V
POWER-IN ORING
AP3101A
Q2
Q1B
LBSS84DW1T1G
Q1A
LBSS84DW1T1G
R12
100K/1%
R14
100K/1%
Q4
Q3B
LBSS84DW1T1G
Q3A
LBSS84DW1T1G
R21
100K/1%
R23
100K/1%
SiPEED
Sheet: /POWER/
File: SYS_POWER.kicad_sch
Title: Tang_Mega_60K_Dock(NEO)
Size: A4
Date: 2025-03-31
Rev: 1.4
KiCad E.D.A. 8.0.6
Id: 4/21

IIC connection



USB connection

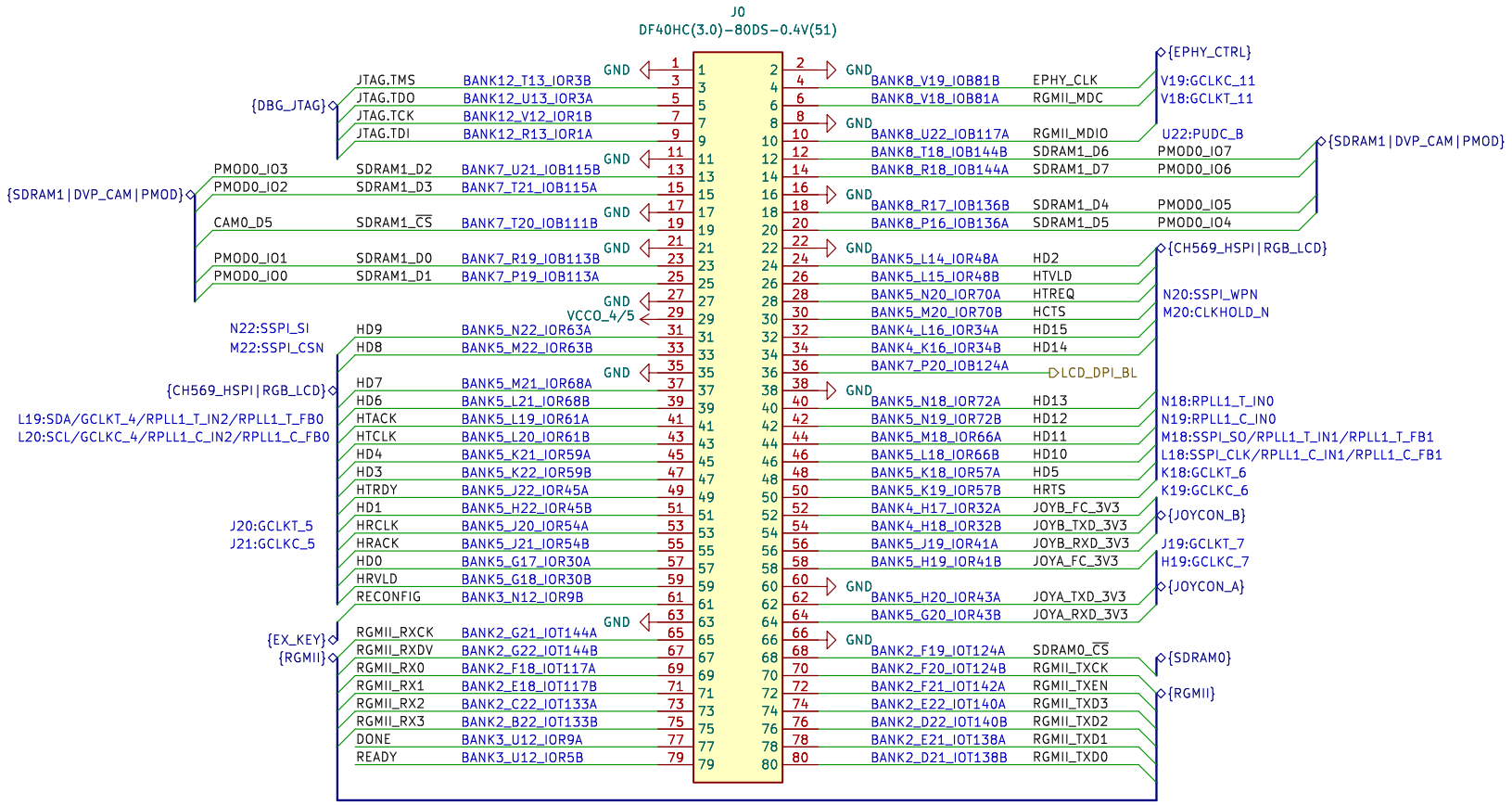


SiPEED

Sheet: /Block_Diagram/
File: Block_Diagram.kicad_sch

Title: Tang_Mega_60K_Dock(NEO)

Size: A4	Date: 2025-03-31	**Rev: 1.4**
KiCad E.D.A. 8.0.6		Id: 5/21



H7_LU1 MountingHole_Pad — GND
H7_RU1 MountingHole_Pad — GND
H7_LD1 MountingHole_Pad — GND
H7_RD1 MountingHole_Pad — GND

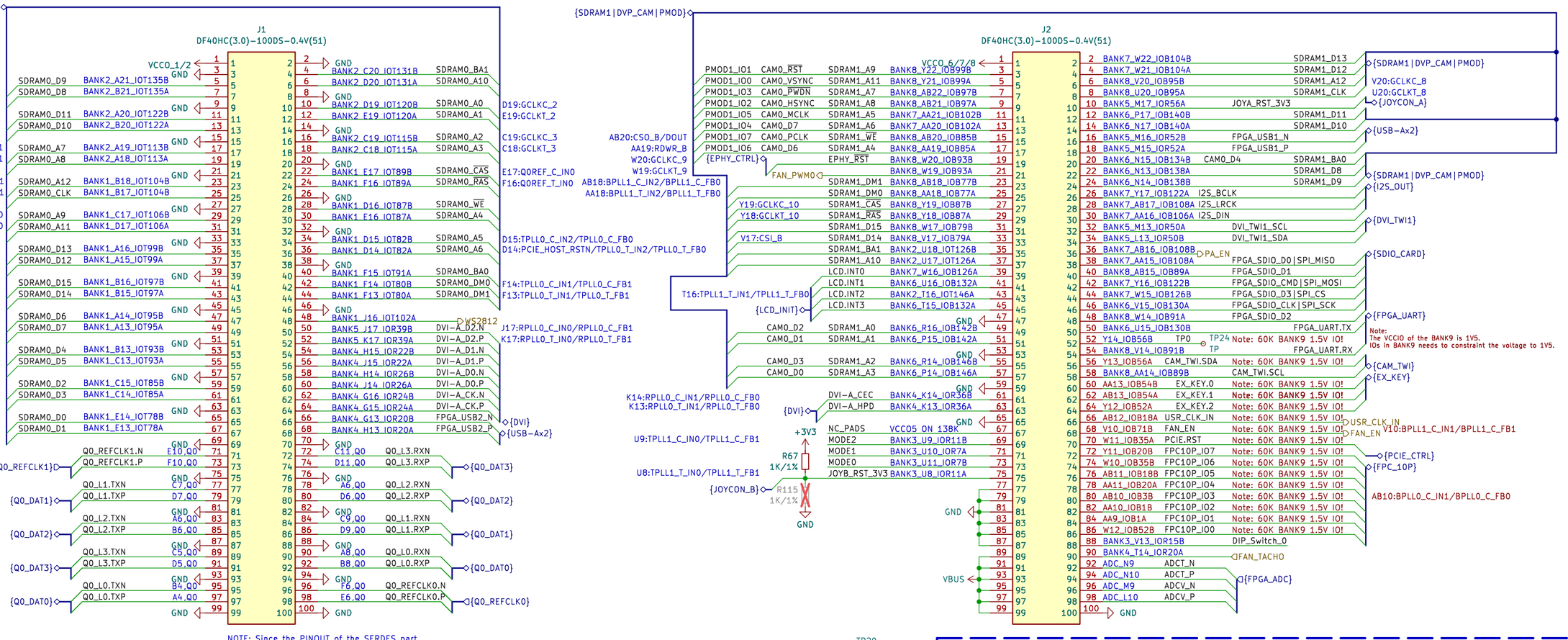
H1 M2.5xΦ4x3.0+1.6Hx2.4 — GND
H2 M2.5xΦ4x3.0+1.6Hx2.4 — GND
H3 M2.5xΦ4x3.0+1.6Hx2.4 — GND
H4 M2.5xΦ4x3.0+1.6Hx2.4 — GND
H5 MH_HEATSINK
H6 MH_HEATSINK

SiPEED

Sheet: /SOM_BTB0/
File: SOM_BTB0.kicad_sch

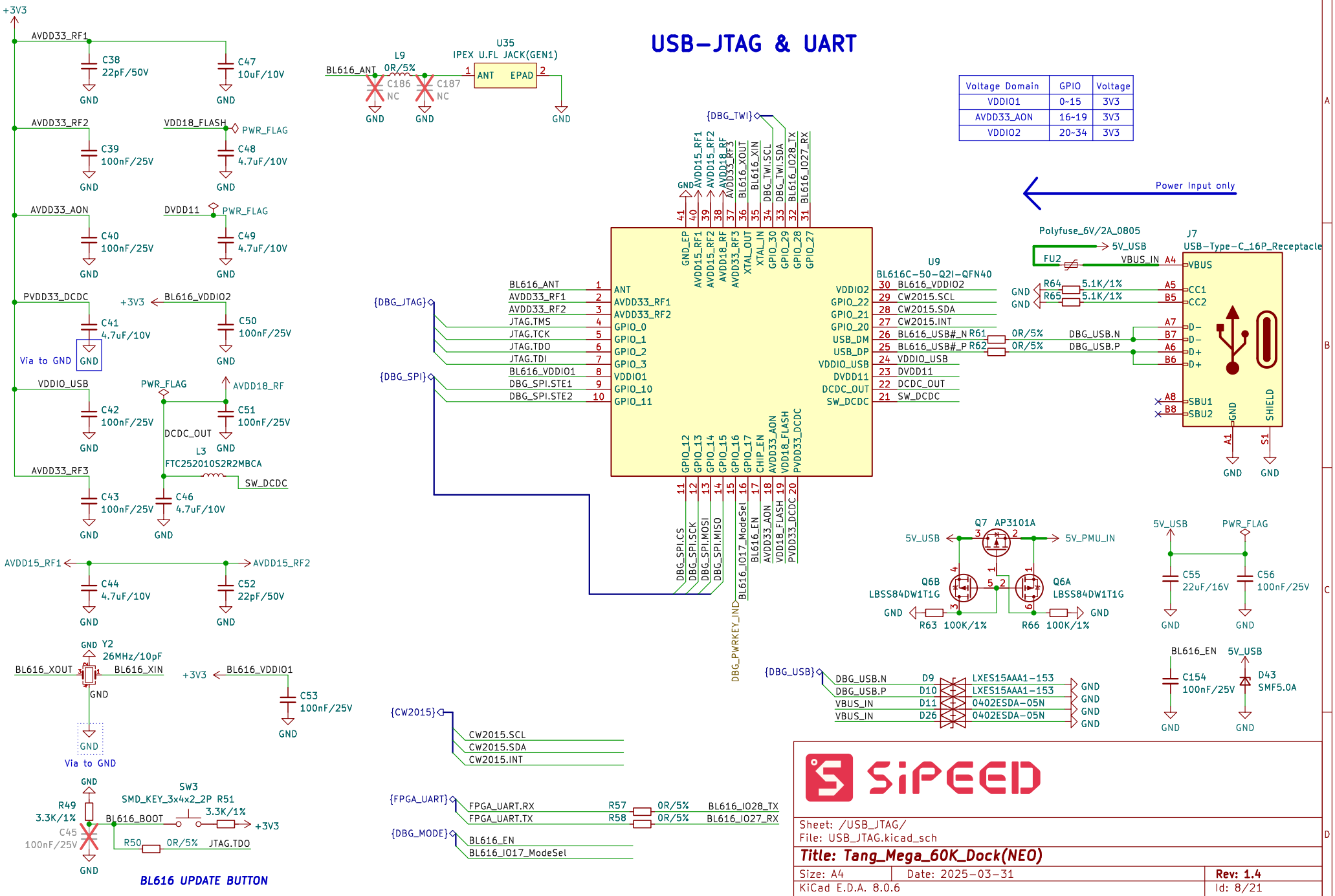
Title: Tang_Mega_60K_Dock(NEO)

Size: A4	Date: 2025-03-31	Rev: 1.4
KiCad E.D.A. 8.0.6		Id: 6/21


{SDRAM0}
J1
DF40HC(3.0)-100DS-0.4V(51)
VCCO_1/2
GND
SDRAM0_D9 BANK2_A21_IOT135B
SDRAM0_D8 BANK2_B21_IOT135A
SDRAM0_D11 BANK2_A20_IOT122B
SDRAM0_D10 BANK2_B20_IOT122A
SDRAM0_A7 BANK2_A19_IOT113B
SDRAM0_A8 BANK2_A18_IOT113A
SDRAM0_A12 BANK1_B18_IOT104B
SDRAM0_CLK BANK1_B17_IOT104B
SDRAM0_A9 BANK1_C17_IOT106B
SDRAM0_A11 BANK1_D17_IOT106A
SDRAM0_D13 BANK1_A16_IOT99B
SDRAM0_D12 BANK1_A15_IOT99A
SDRAM0_D15 BANK1_B16_IOT97B
SDRAM0_D14 BANK1_B15_IOT97A
SDRAM0_D6 BANK1_A14_IOT95B
SDRAM0_D7 BANK1_A13_IOT95A
SDRAM0_D4 BANK1_B13_IOT93B
SDRAM0_D5 BANK1_C13_IOT93A
SDRAM0_D2 BANK1_C15_IOT85B
SDRAM0_D3 BANK1_C14_IOT85A
SDRAM0_D0 BANK1_E14_IOT78B
SDRAM0_D1 BANK1_E13_IOT78A
A19:QOREF_C_IN1
A18:QOREF_T_IN1
B18:GCLKC_1
B17:GCLKT_1
C17:GCLKC_0
D17:GCLKT_0
BANK2_C20_IOT131B SDRAM0_BA1
BANK2_D20_IOT131A SDRAM0_A10
BANK2_D19_IOT120B SDRAM0_A0
BANK2_E19_IOT120A SDRAM0_A1
BANK2_C19_IOT115B SDRAM0_A2
BANK2_C18_IOT115A SDRAM0_A3
BANK1_E17_IOT89B SDRAM0_CAS
BANK1_F16_IOT89A SDRAM0_RAS
BANK1_D16_IOT87B SDRAM0_WE
BANK1_E16_IOT87A SDRAM0_A4
BANK1_D15_IOT82B SDRAM0_A5
BANK1_D14_IOT82A SDRAM0_A6
BANK1_F15_IOT91A SDRAM0_BA0
BANK1_F14_IOT80B SDRAM0_DM0
BANK1_F13_IOT80A SDRAM0_DM1
BANK1_J16_IOT102A WS2812
BANK5_J17_IOR39B DVI-A_D2.N
BANK5_K17_IOR39A DVI-A_D2.P
BANK4_H15_IOR22B DVI-A_D1.N
BANK4_J15_IOR22A DVI-A_D1.P
BANK4_H14_IOR26B DVI-A_D0.N
BANK4_J14_IOR26A DVI-A_D0.P
BANK4_G16_IOR24B DVI-A_CK.N
BANK4_G15_IOR24A DVI-A_CK.P
BANK4_G13_IOR20B FPGA_USB2_N
BANK4_H13_IOR20A FPGA_USB2_P
D19:GCLKC_2
E19:GCLKT_2
C19:GCLKC_3
C18:GCLKT_3
E17:QOREF_C_IN0
F16:QOREF_T_IN0
D15:TPLL0_C_IN2/TPLL0_C_FB0
D14:PCIE_HOST_RSTN/TPLL0_T_IN2/TPLL0_T_FB0
F14:TPLL0_C_IN1/TPLL0_C_FB1
F13:TPLL0_T_IN1/TPLL0_T_FB1
J17:RPLL0_C_IN0/RPLL0_C_FB1
K17:RPLL0_T_IN0/RPLL0_T_FB1
{DVI}
{USB-Ax2}
{Q0_REFCLK1}
Q0_REFCLK1.N E10 Q0
Q0_REFCLK1.P F10 Q0
Q0_L1.TXN C7 Q0
Q0_L1.TXP D7 Q0
{Q0_DAT1}
Q0_L2.TXN A6 Q0
Q0_L2.TXP B6 Q0
{Q0_DAT2}
Q0_L3.TXN C5 Q0
Q0_L3.TXP D5 Q0
{Q0_DAT3}
Q0_L0.TXN B4 Q0
Q0_L0.TXP A4 Q0
{Q0_DAT0}
C11 Q0 Q0_L3.RXN
D11 Q0 Q0_L3.RXP
A8 Q0 Q0_L2.RXN
D6 Q0 Q0_L2.RXP
C9 Q0 Q0_L1.RXN
D9 Q0 Q0_L1.RXP
A8 Q0 Q0_L0.RXN
B8 Q0 Q0_L0.RXP
F6 Q0 Q0_REFCLK0.N
E6 Q0 Q0_REFCLK0.P
{Q0_DAT3}
{Q0_DAT2}
{Q0_DAT1}
{Q0_DAT0}
{Q0_REFCLK0}
NOTE: Since the PINOUT of the SERDES part
is not completely consistent with 138K,
the Quad0 lanes 1 and 3 here are swapped
relative to 138K, so PCIe can only work
in x1 mode when SOM is 60K!!!
{SDRAM1|DVP_CAM|PMOD}
J2
DF40HC(3.0)-100DS-0.4V(51)
VCCO_6/7/8
PMOD1_IO1 CAMO_RST SDRAM1_A9 BANK8_Y22_IOB99B
PMOD1_IO0 CAMO_VSYNC SDRAM1_A11 BANK8_Y21_IOB99A
PMOD1_IO3 CAMO_PWDN SDRAM1_A7 BANK8_AB22_IOB97B
PMOD1_IO2 CAMO_HSYNC SDRAM1_A8 BANK8_AB21_IOB97A
PMOD1_IO5 CAMO_MCLK SDRAM1_A5 BANK7_AA21_IOB102B
PMOD1_IO4 CAMO_D7 SDRAM1_A6 BANK7_AA20_IOB102A
PMOD1_IO7 CAMO_PCLK SDRAM1_WE BANK8_AB20_IOB85B
PMOD1_IO6 CAMO_D6 SDRAM1_A4 BANK8_AA19_IOB85A
{EPHY_CTRL} EPHY_RST BANK8_W20_IOB93B
FAN_PWM0 BANK8_W19_IOB93A
SDRAM1_DM1 BANK8_AB18_IOB77B
SDRAM1_DM0 BANK8_AA18_IOB77A
SDRAM1_CAS BANK8_Y19_IOB87B
SDRAM1_RAS BANK8_Y18_IOB87A
SDRAM1_D15 BANK8_W17_IOB79B
SDRAM1_D14 BANK8_V17_IOB79A
SDRAM1_BA1 BANK2_U18_IOT126B
SDRAM1_A10 BANK2_U17_IOT126A
LCD.INT0 BANK7_W16_IOB126A
LCD.INT1 BANK6_U16_IOB132A
LCD.INT2 BANK2_T16_IOT146A
LCD.INT3 BANK6_T15_IOB132A
{LCD_INIT}
CAMO_D2 SDRAM1_A0 BANK6_R16_IOB142B
CAMO_D1 SDRAM1_A1 BANK6_P15_IOB142A
CAMO_D3 SDRAM1_A2 BANK6_R14_IOB146B
CAMO_D0 SDRAM1_A3 BANK6_P14_IOB146A
DVI-A_CEC BANK4_K14_IOR36B
DVI-A_HPD BANK4_K13_IOR36A
NC_PADS VCCO5 ON 138K
MODE2 BANK3_U9_IOR11B
MODE1 BANK3_U10_IOR7A
MODE0 BANK3_U11_IOR7B
JOYB_RST_3V3 BANK3_U8_IOR11A
BANK7_W22_IOB104B SDRAM1_D13
BANK7_W21_IOB104A SDRAM1_D12
BANK8_V20_IOB95B SDRAM1_A12
BANK8_U20_IOB95A SDRAM1_CLK
BANK5_M17_IOR56A JOYA_RST_3V3
BANK6_P17_IOB140B SDRAM1_D11
BANK6_N17_IOB140A SDRAM1_D10
BANK5_M16_IOR52B FPGA_USB1_N
BANK5_M15_IOR52A FPGA_USB1_P
BANK6_N15_IOB134B CAMO_D4 SDRAM1_BA0
BANK6_N13_IOB138A SDRAM1_D8
BANK6_N14_IOB138B SDRAM1_D9
BANK7_Y17_IOB122A I2S_BCLK
BANK7_AB17_IOB108A I2S_LRCK
BANK7_AA16_IOB106A I2S_DIN
BANK5_M13_IOR50A DVI_TWI1_SCL
BANK5_L13_IOR50B DVI_TWI1_SDA
BANK7_AB16_IOB108B PA_EN
BANK7_AA15_IOB108A FPGA_SDIO_D0|SPI_MISO
BANK8_AB15_IOB89A FPGA_SDIO_D1
BANK7_Y16_IOB122B FPGA_SDIO_CMD|SPI_MOSI
BANK7_W15_IOB126B FPGA_SDIO_D3|SPI_CS
BANK6_V15_IOB130A FPGA_SDIO_CLK|SPI_SCK
BANK8_W14_IOB91A FPGA_SDIO_D2
BANK6_U15_IOB130B FPGA_UART.TX
Y14_IOB56B TPO TP24
BANK8_V14_IOB91B TP FPGA_UART.RX
Y13_IOB56A CAM_TWI.SDA
BANK8_AA14_IOB89B CAM_TWI.SCL
AA13_IOB54B EX_KEY.0
AB13_IOB54A EX_KEY.1
Y12_IOB52A EX_KEY.2
AB12_IOB18A USR_CLK_IN
V10_IOB71B FAN_EN
W11_IOB35A PCIE.RST
Y11_IOB20B FPC10P_IO7
W10_IOB35B FPC10P_IO6
AB11_IOB18B FPC10P_IO5
AA11_IOB20A FPC10P_IO4
AB10_IOB3B FPC10P_IO3
AA10_IOB1B FPC10P_IO2
AA9_IOB1A FPC10P_IO1
W12_IOB52B FPC10P_IO0
BANK3_V13_IOR15B DIP_Switch_0
BANK4_T14_IOR20A FAN_TACHO
ADC_N9 ADCT_N
ADC_N10 ADCT_P
ADC_M9 ADCV_N
ADC_L10 ADCV_P
{FPGA_ADC}
Note: 60K BANK9 1.5V IO!
Note:
The VCCIO of the BANK9 is 1V5.
IOs in BANK9 needs to constraint the voltage to 1V5.
{SDRAM1|DVP_CAM|PMOD}
V20:GCLKC_8
U20:GCLKT_8
{JOYCON_A}
{USB-Ax2}
{SDRAM1|DVP_CAM|PMOD}
{I2S_OUT}
{DVI_TWI1}
{SDIO_CARD}
{FPGA_UART}
{CAM_TWI}
{EX_KEY}
USR_CLK_IN
V10:BPLL1_C_IN1/BPLL1_C_FB1
FAN_EN
{PCIE_CTRL}
{FPC_10P}
AB10:BPLL0_C_IN1/BPLL0_C_FB0
AB20:CSO_B/DOUT
AA19:RDWR_B
W20:GCLKC_9
W19:GCLKT_9
AB18:BPLL1_C_IN2/BPLL1_C_FB0
AA18:BPLL1_T_IN2/BPLL1_T_FB0
Y19:GCLKC_10
Y18:GCLKT_10
V17:CSI_B
T16:TPLL1_T_IN1/TPLL1_T_FB0
K14:RPLL0_C_IN1/RPLL0_C_FB0
K13:RPLL0_T_IN1/RPLL0_T_FB0
U9:TPLL1_C_IN0/TPLL1_C_FB1
U8:TPLL1_T_IN0/TPLL1_T_FB1
{DVI}
{JOYCON_B}
+3V3
R67 1K/1%
R115 1K/1%
GND
VBUS
VCCO_1/2 TP20 TP
VCCO_4/5 TP21 TP
VCCO_6/7/8 TP22 TP
NC_PADS TP23 TP
GW5AT-60 Configuration Modes
Configuration MODE | MODE[2..0] | Bus Width | Description
JTAG | XXX/101 | – | GW5AT series of FPGA products are configured by external Host via JTAG interface
MSPI | 001 | x1, x2, x4 | As a Master, FPGA reads data from external Flash (or other devices) via the SPI interface for configuration
SSPI | 010 | x1, x4 | GW5AT series of FPGA products are configured by external Host via SPI interface
Master SERIAL | 000 | x1 | Before FPGA used as a Slave FPGA reads data from external devices via the DIN interface for configuration
Slave SERIAL | 111 | x1 | GW5AT series of FPGA products are configured by external Host via DIN interface
Master CPU | 100 | x8, x16 | Before FPGA used as a Slave FPGA reads data from external devices via the DBUS interface for configuration
Slave CPU | 110 | x8, x16 | GW5AT series of FPGA products are configured by external Host via DBUS interface
Please refer to GOWIN Manual UG718/UG718E for more details.
BL616 SEL[0..1] SWITCH
NO. | CONFIG | SEL0(EN) | SEL1(MODE)
0 | Disable BL616 Use other debugger | 0 | X
1 | JTAG mode | 1 | 1
2 | Game mode | 1 | 0
R2 150K/1% BL616_EN
R3 150K/1% BL616_IO17_ModeSel
{DBG_MODE}
Switch
SW2
DIP_KEY_DIP_6P_2.54mm_蓝色
R4 1.5K/5% MODE0
R5 1.5K/5% MODE1
R6 1.5K/5% MODE2
R7 1.5K/5%
BL616_IO17_ModeSel
BL616_EN
{DBG_MODE}
R8 1K/1%
R43 1K/1%
R44 1K/1%
DIP_Switch_0
R53 51K/1%
R54 51K/1%
+3V3
DIP_Switch_0 : FPGA internal pull-up to 3.3V
SiPEED
Sheet: /SOM_BTB1&2/
File: SOM_BTB1-2.kicad_sch
Title: Tang_Mega_60K_Dock(NEO)
Size: A3
Date: 2025-03-31
Rev: 1.4
KiCad E.D.A. 8.0.6
Id: 7/21

USB-JTAG & UART

Voltage Domain	GPIO	Voltage
VDDIO1	0-15	3V3
AVDD33_AON	16-19	3V3
VDDIO2	20-34	3V3



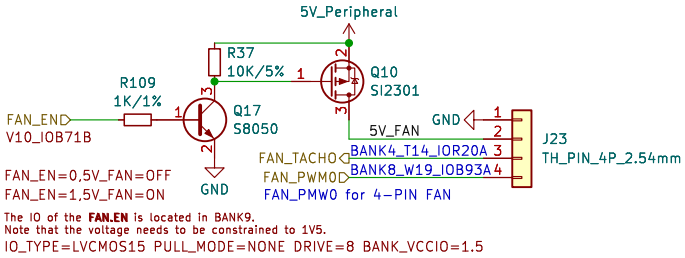
SiPEED

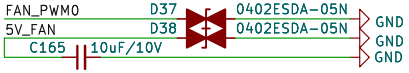
Sheet: /USB_JTAG/
File: USB_JTAG.kicad_sch

Title: Tang_Mega_60K_Dock(NEO)

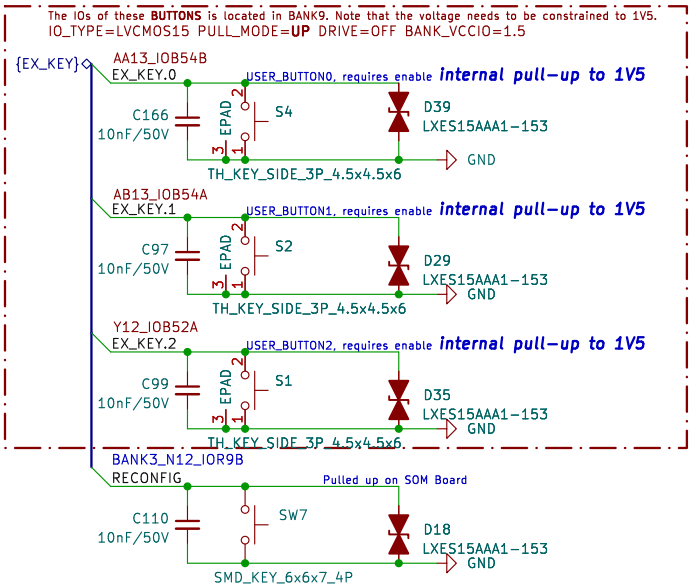
Size: A4	Date: 2025-03-31	Rev: 1.4
KiCad E.D.A. 8.0.6		Id: 8/21

FAN Connector

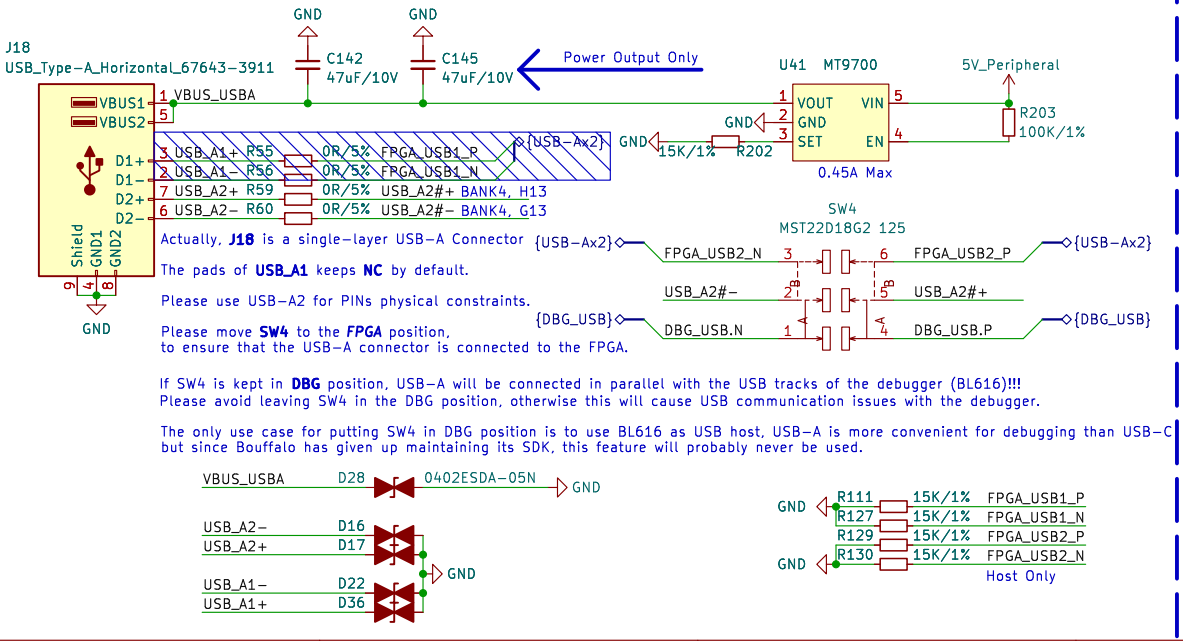




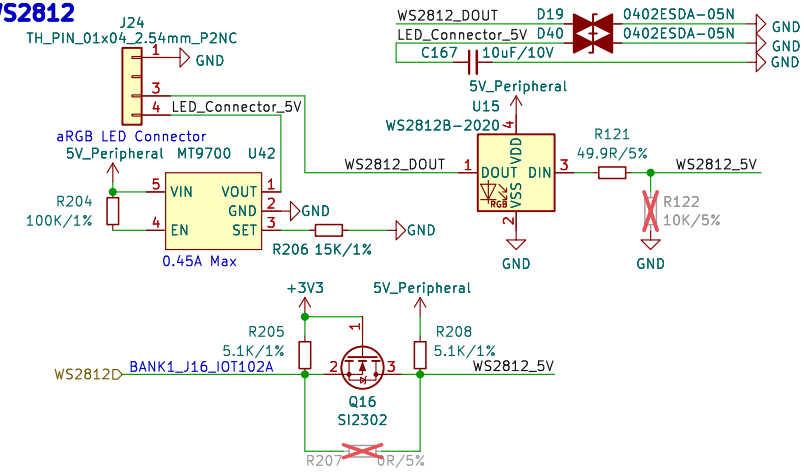
BUTTON



USB_A_x2



WS2812



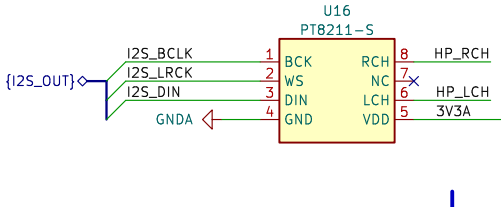
SiPEED

Sheet: /PERIPHERALS/	
File: PERIPHERALS.kicad_sch	
Title: Tang_Mega_60K_Dock(NEO)	
Size: A4	Date: 2025-03-31
KiCad E.D.A. 8.0.6	Id: 9/21

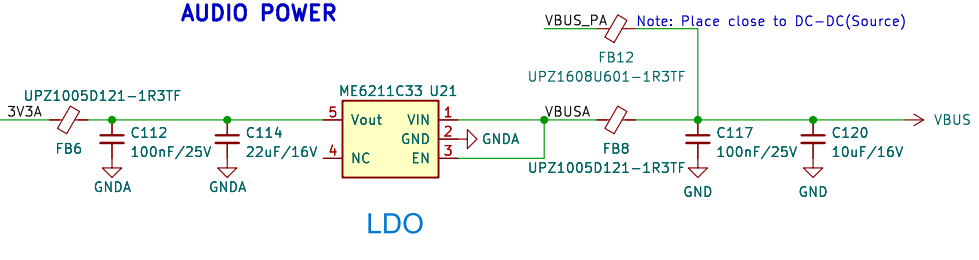
Rev: 1.4

FPGA-Audio

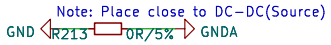
STEREO DAC



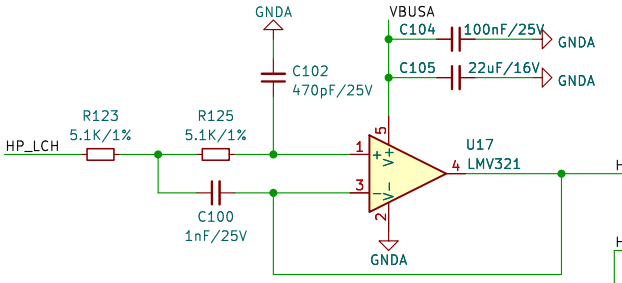
AUDIO POWER

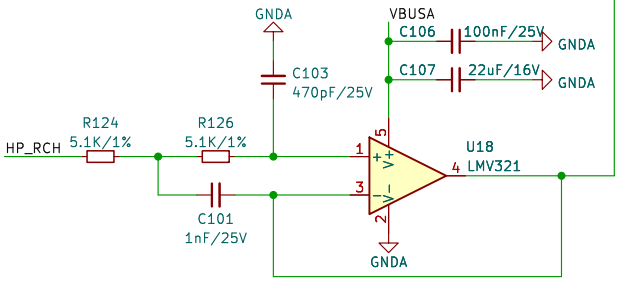


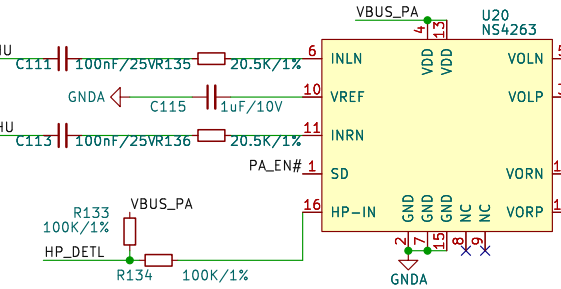
LDO



LOW-PASS FILTER

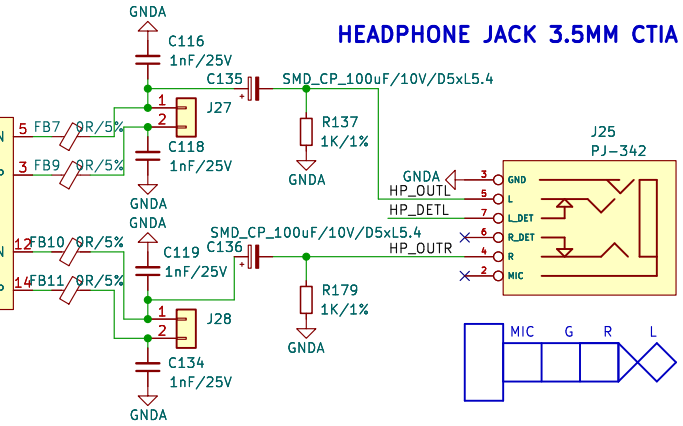






Note:
PA_EN=1 : PA OFF
PA_EN=0 : PA ON

HEADPHONE JACK 3.5MM CTIA



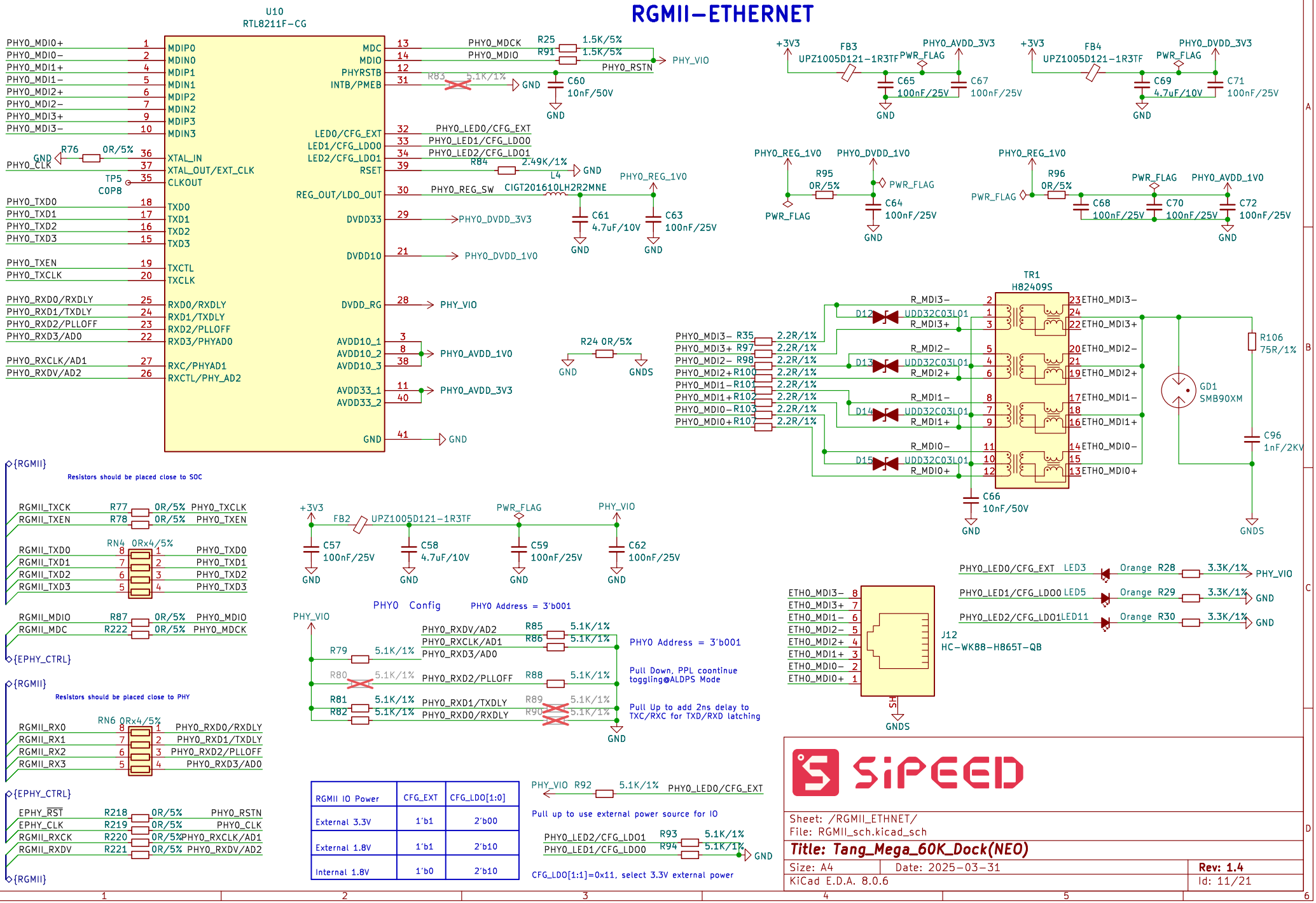
When the jack is unpluged, L/R_DET is connected to L/R.
HP-IN = 0
Speaker enable

When the jack is pluged, L/R_DET is disconnected to L/R.
HP-IN = 1
Headphone enable

SiPEED

Sheet: /FPGA_AUDIO/		
File: DAC+PA.kicad_sch		
Title: Tang_Mega_60K_Dock(NEO)		
Size: A4	Date: 2025-03-31	**Rev: 1.4**
KiCad E.D.A. 8.0.6		Id: 10/21

RGMII—ETHERNET



SiPEED

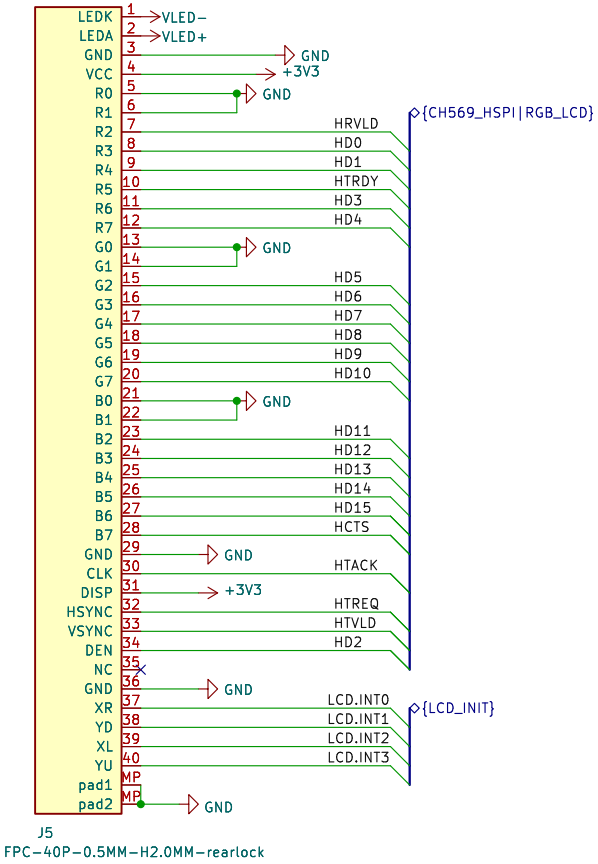
Sheet: /RGMII_ETHNET/
File: RGMII_sch.kicad_sch

Title: Tang_Mega_60K_Dock(NEO)

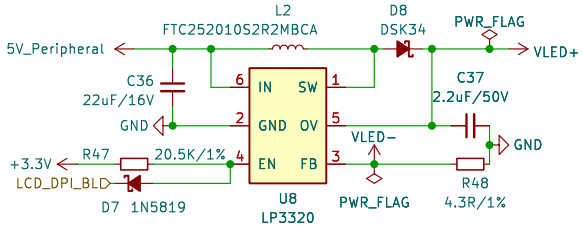
Size: A4	Date: 2025-03-31	**Rev: 1.4**
KiCad E.D.A. 8.0.6		Id: 11/21

RGB LCD Display

RGB LCD Display Connector

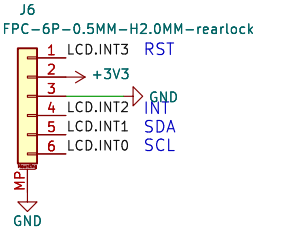


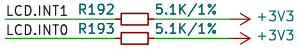
BACKLIGHT Driver



Feedback Voltage = 250mV
BackLight Current = 250mV/4.3R ≈ 58.1mA
7inch LCD QTM070TFT40-001 : IF=20*3=60mA

Capacitive Touch Connector





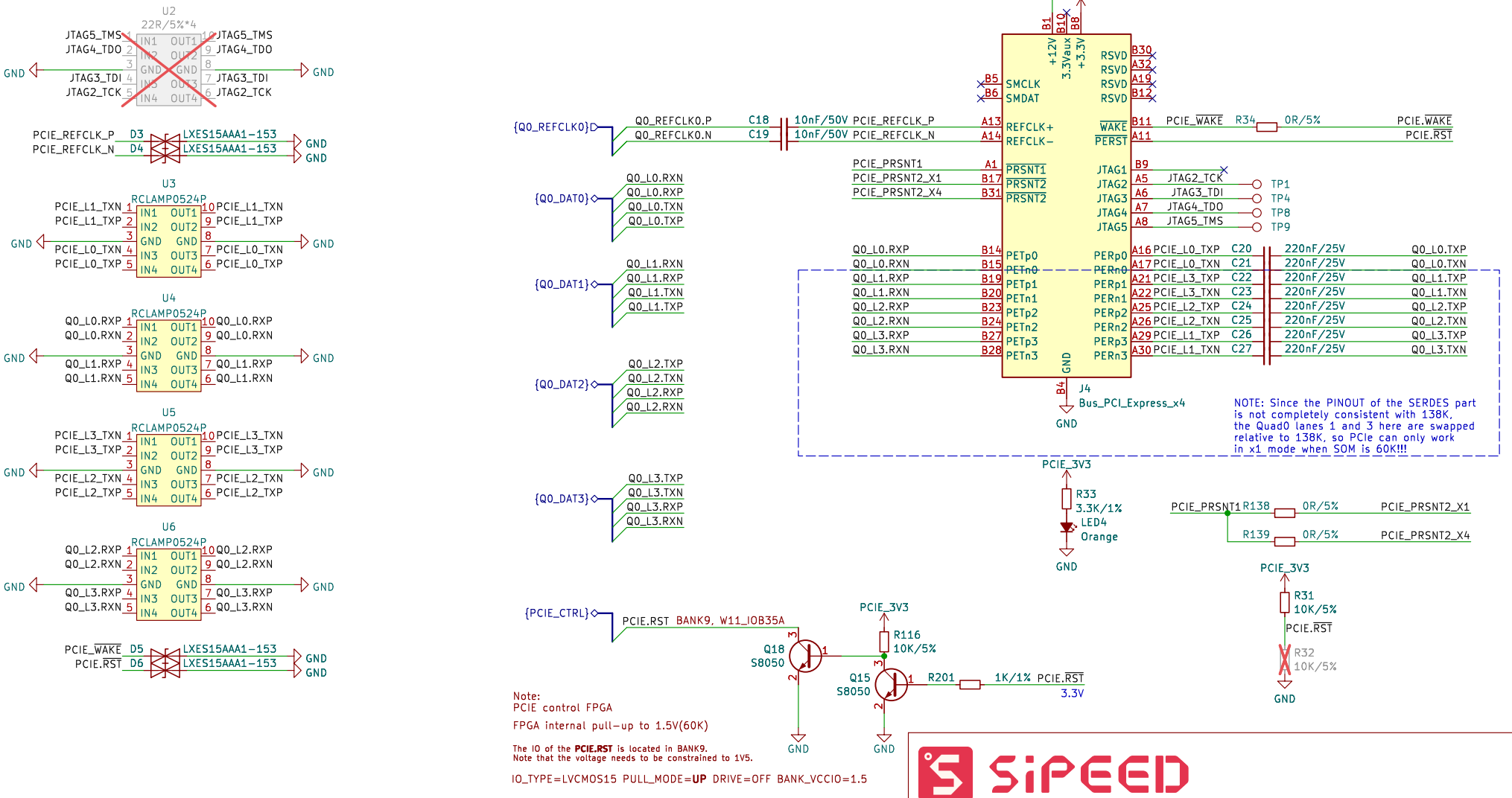
SiPEED

Sheet: /SYS_DPI/
File: LCD_DPI.kicad_sch

Title: Tang_Mega_60K_Dock(NEO)

Size: A4	Date: 2025-03-31	**Rev: 1.4**
KiCad E.D.A. 8.0.6		Id: 12/21

PCI-Express X4

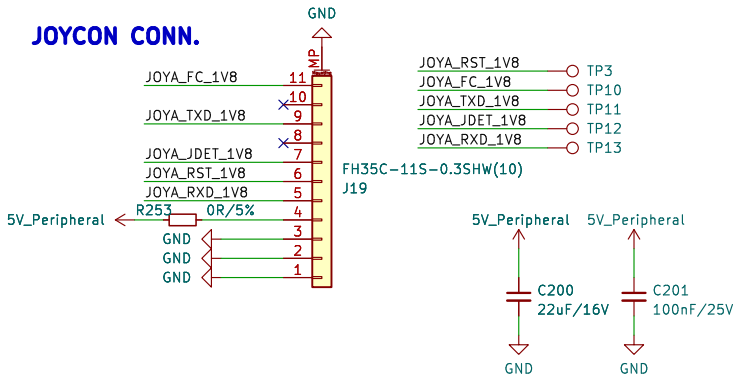


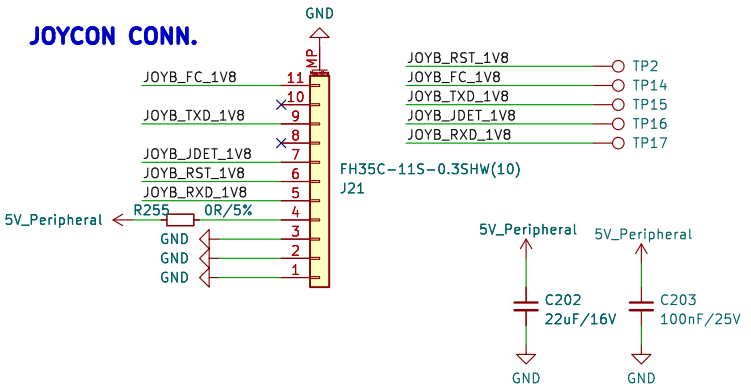
SiPEED

Sheet: /PCIE_GT/
File: SERDES_PCIE.kicad_sch

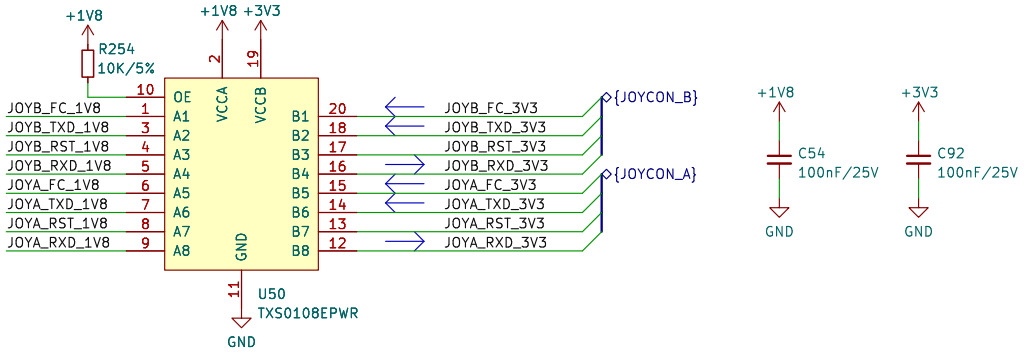
Title: Tang_Mega_60K_Dock(NEO)

Size: A4	Date: 2025-03-31	**Rev: 1.4**
KiCad E.D.A. 8.0.6		Id: 13/21





Bi-directional Voltage Translator



Note:
RST=0: Enable JOYCON
RST=1: Disable JOYCON

JDET=1:
HIGH when connected to console via Bluetooth.
Joy-Con will not send serial data post-handshake unless pin is pulled LOW by console.

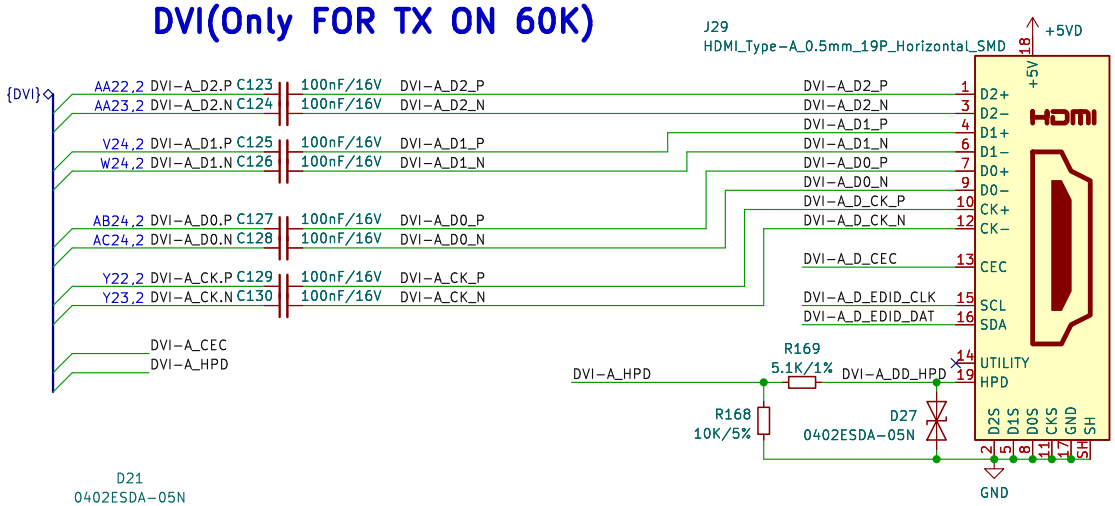
SiPEED

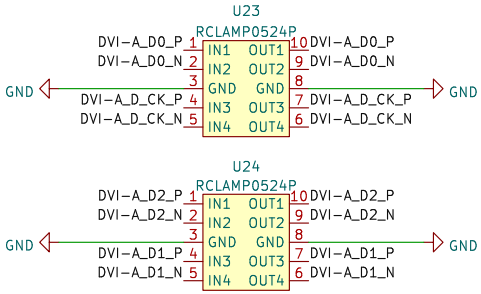
Sheet: /JOYCON/
File: JOYCON.kicad_sch

Title: Tang_Mega_60K_Dock(NEO)

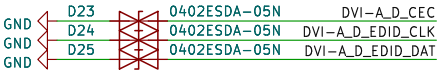
Size: A4	Date: 2025-03-31	**Rev: 1.4**
KiCad E.D.A. 8.0.6		Id: 14/21

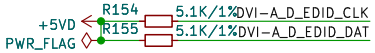
DVI(Only FOR TX ON 60K)

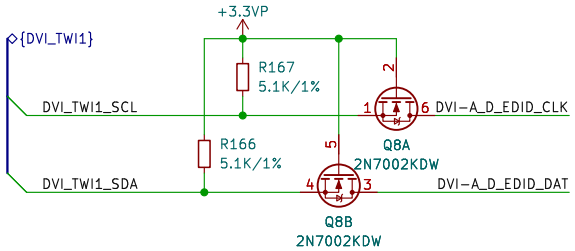


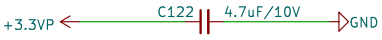






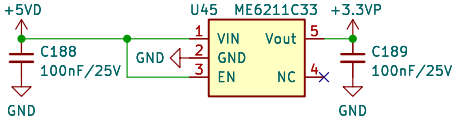


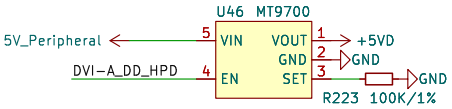




DVI Power control

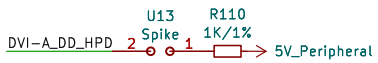
(Save Battery Power)

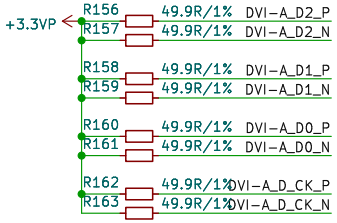


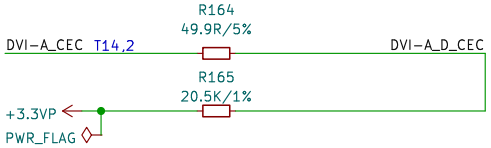


Insert HDMI : HPD=1
Unplug HDMI : HPD=0

Current limit: 68mA







SiPEED

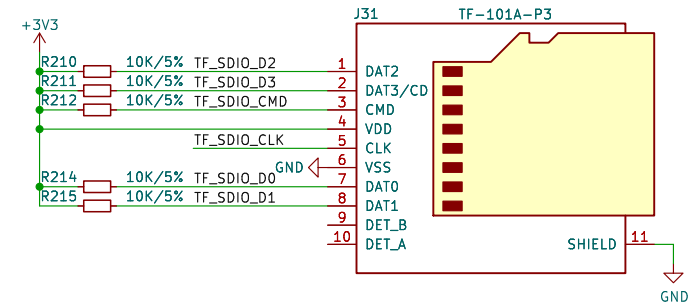
Sheet: /FPGA_TMDS_A/
File: DVI_A.kicad_sch

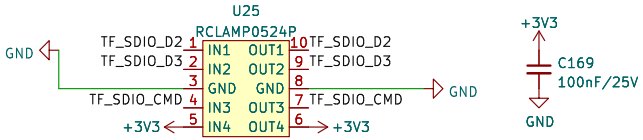
Title: Tang_Mega_60K_Dock(NEO)

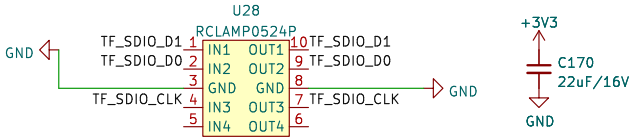
Size: A4	Date: 2025-03-31	**Rev: 1.4**
KiCad E.D.A. 8.0.6		Id: 15/21

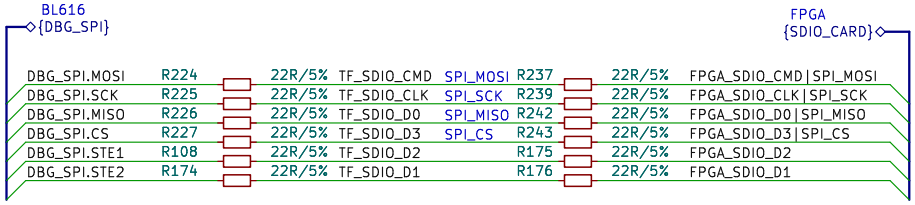
STORAGE

BL616 MICRO SD CARD





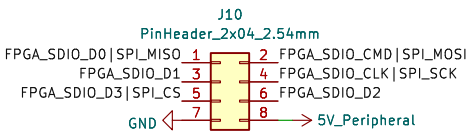




EVB mode : BL616_SPI high-Z , FPGA can work in SPI mode or SDIO mode

Game mode : BL616_SPI Master , FPGA_SPI Slave , FPGA_SPI_CS diasble , FPGA_SDIO_D1/D2 work as SPI_CS(input) or other function

Q1 : How to switch between EVB mode and Game mode?
A1: Switch DBG_ModeSel through SW2(6P Switch DIP)



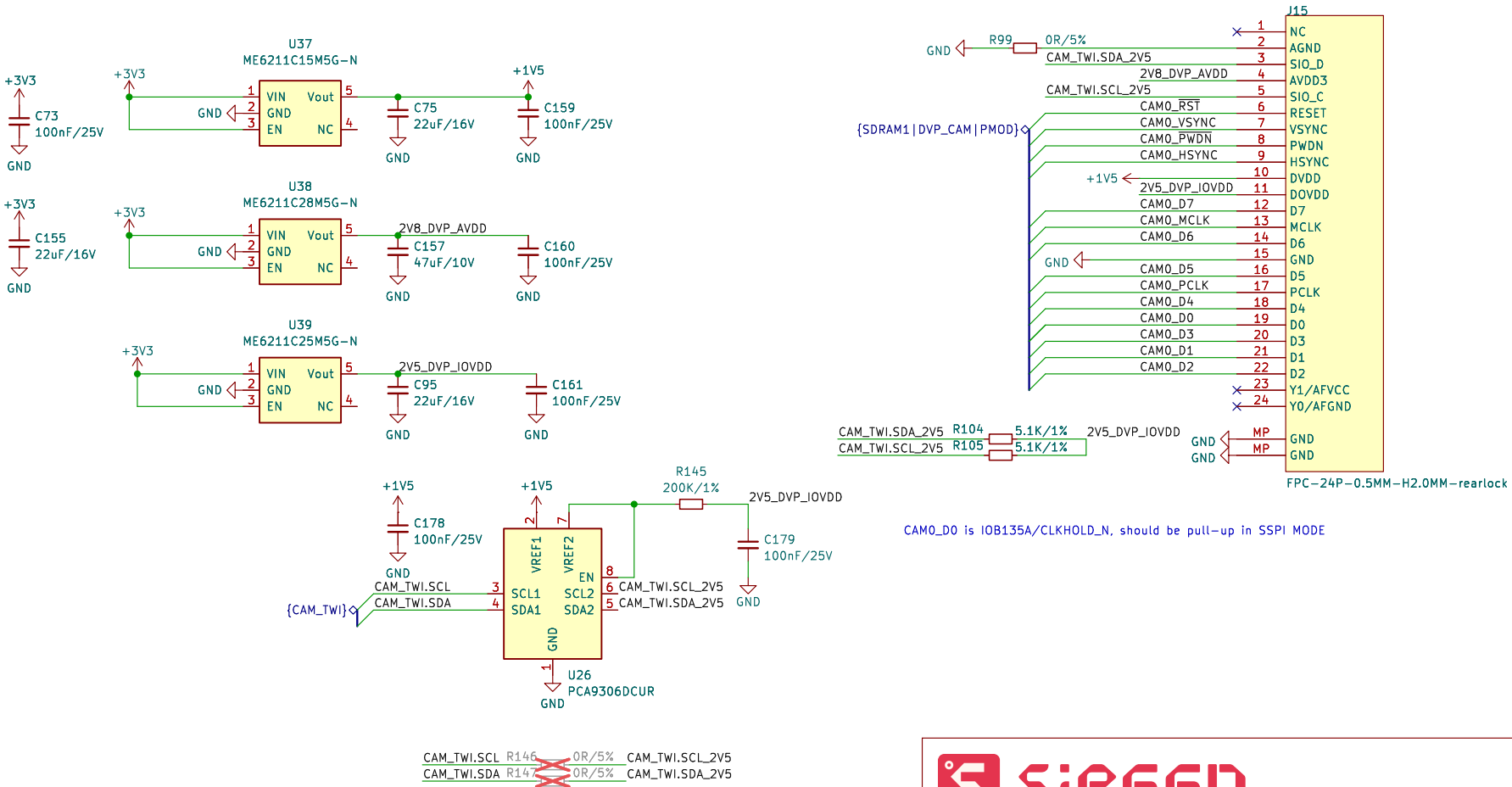
SiPEED

Sheet: /SYS_STORAGE/
File: SDIO&M.2.kicad_sch

Title: Tang_Mega_60K_Dock(NEO)

Size: A4	Date: 2025-03-31	**Rev: 1.4**
KiCad E.D.A. 8.0.6		Id: 16/21

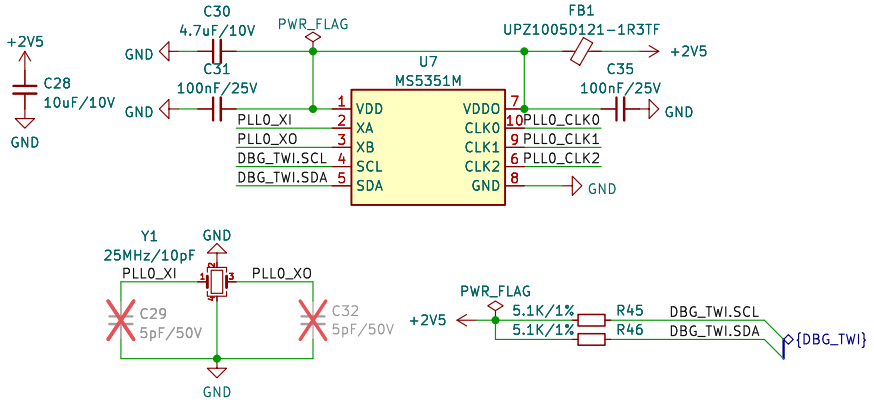
DVP-CAMERA



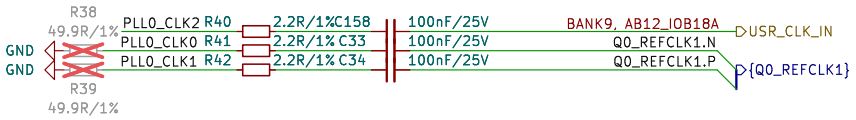
Sheet: /DVP_CAMERA/		
File: DVP_CAM.kicad_sch		
Title: Tang_Mega_60K_Dock(NEO)		
Size: A4	Date: 2025-03-31	**Rev: 1.4**
KiCad E.D.A. 8.0.6		Id: 17/21

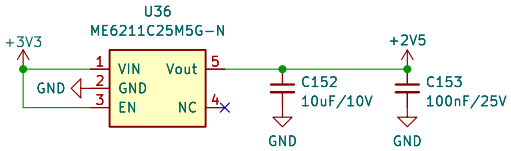
CLOCK

EXT. PLL

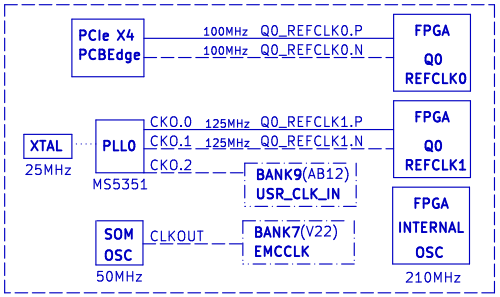


The IO of **USR_CLK_IN** is located in BANK9. Note that the voltage needs to be constrained to 1V5.
IO_TYPE=LVCMOS15 PULL_MODE=DOWN DRIVE=OFF BANK_VCCIO=1.5





COLCK TREE



FPGA internal clock generator OSC.
The user can change the value of the option "Frequency Divider" to generate a different output clock frequency, which can be one of 64 values: 1/2, 1/3, 1/4, 1/6, 1/8, 1/10, ... or 1/126 of the oscillator frequency.

SiPEED

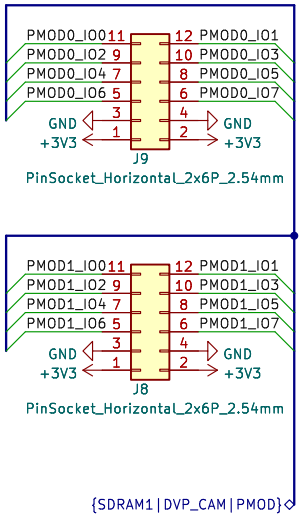
Sheet: /SYS_CLOCK/
File: CLOCK.kicad_sch

Title: Tang_Mega_60K_Dock(NEO)

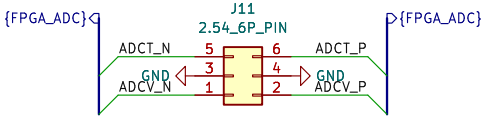
Size: A4	Date: 2025-03-31	**Rev: 1.4**
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EX CONN.

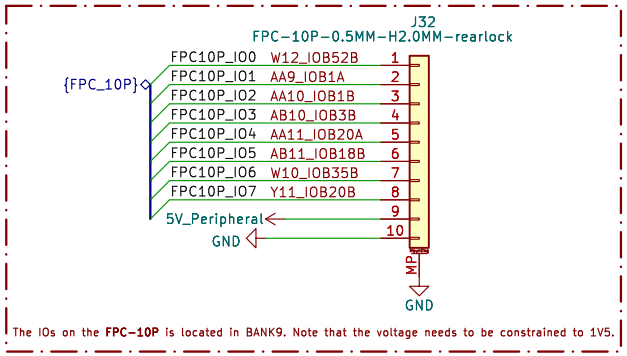
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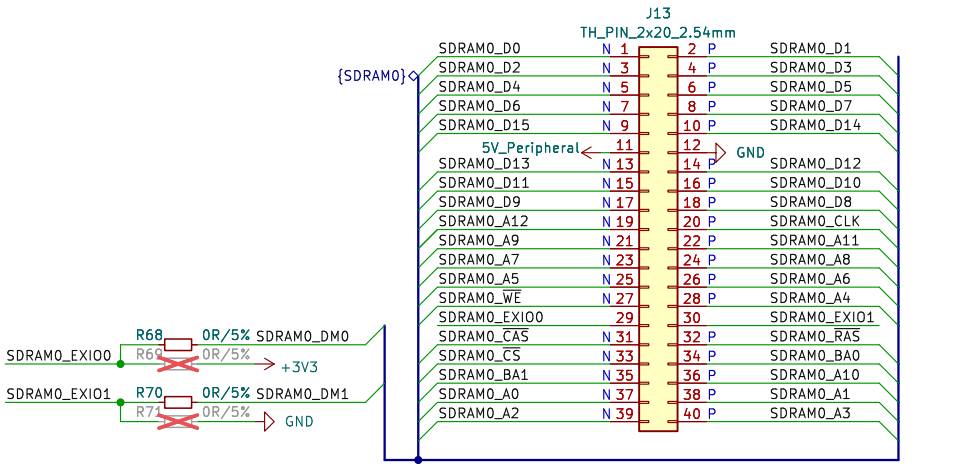
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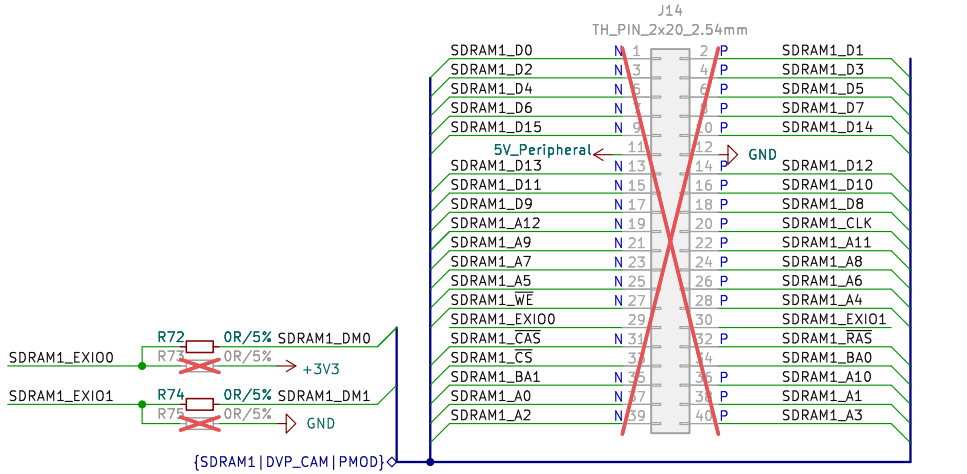
FPC10P CONN.



SDRAM0 CONN. Close to PCIE connector



SDRAM1 CONN. Close to PMOD connector



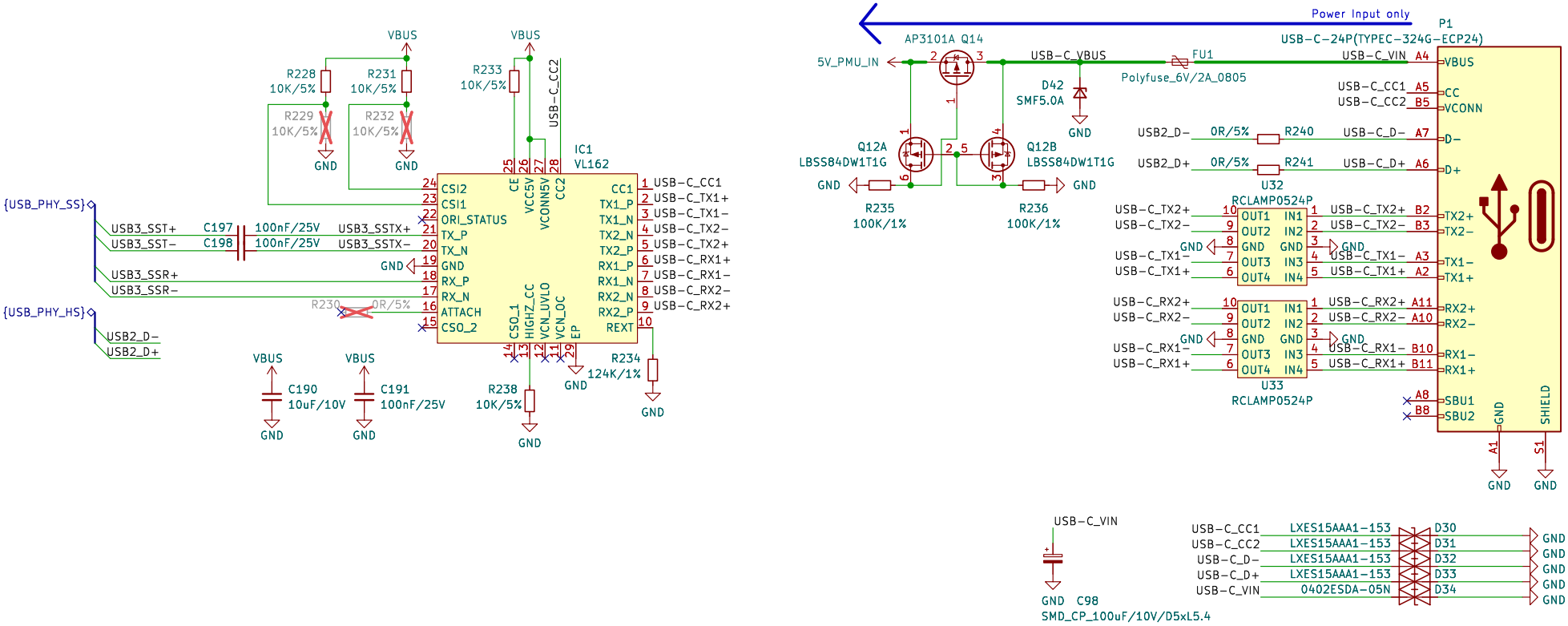
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CH569 ⟷ USB 2:1 Data Switch ⟷ CH569 USB-C Connector



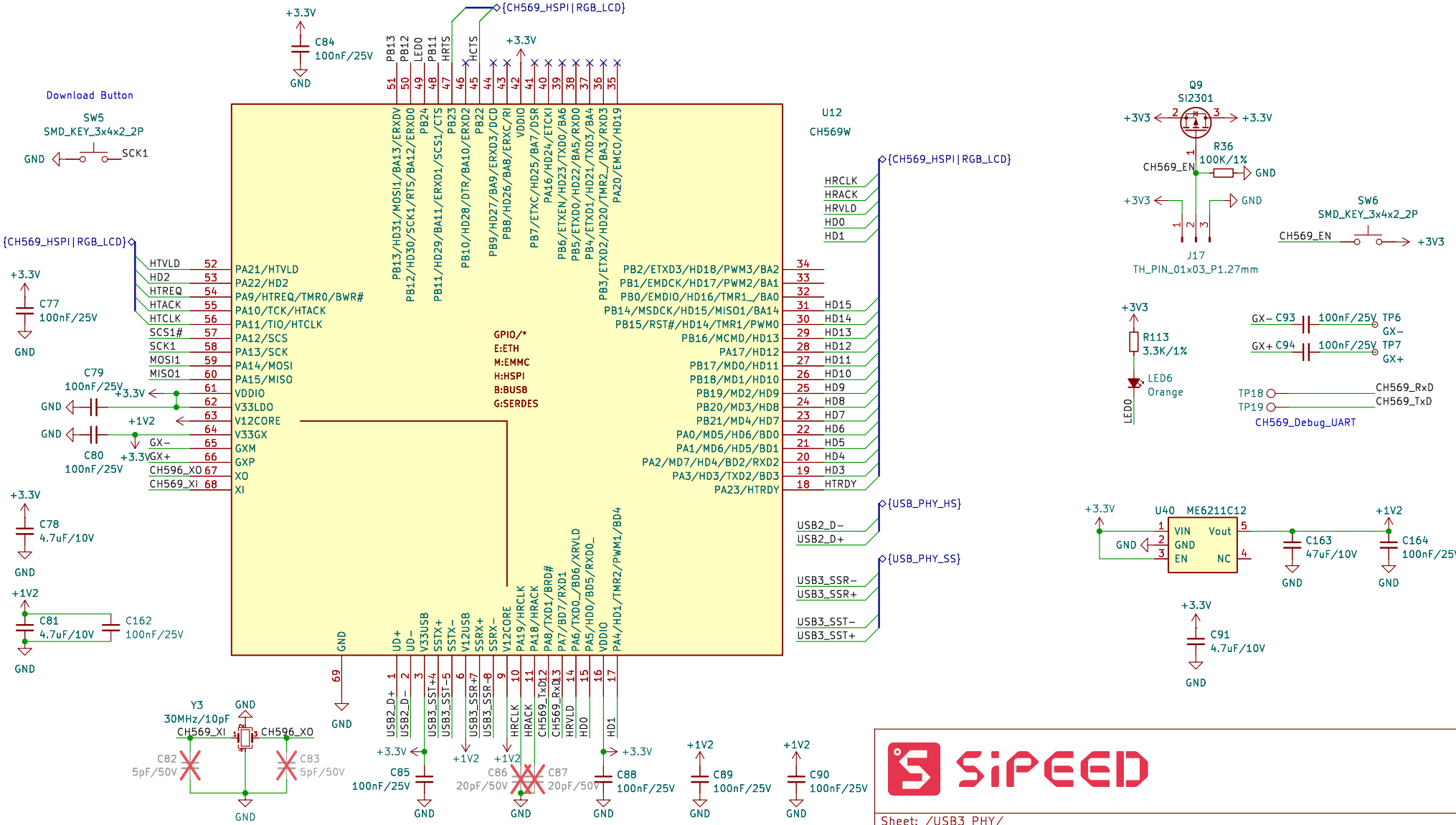
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USB3-PHY



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KiCad E.D.A. 8.0.6	Id: 21/21

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